HW1.

2.1(a). $\Omega_A = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} d\phi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} d\theta \sin\theta = \frac{\pi}{12} \times \cos\theta \Big|_{\frac{\pi}{3}}^{\frac{\pi}{6}} = 0.09582$ sterads.

exactly: $= 0.09582 \times \left(\frac{180}{\pi}\right)^2 = 314.56$ degrees

approximate: $\Omega_A = \Delta\theta \cdot \Delta\phi = \left(\frac{\pi}{3} - \frac{\pi}{4}\right)\left(\frac{\pi}{3} - \frac{\pi}{6}\right)$

$= 0.1371$ sterads

$= 0.1371 \times \left(\frac{180}{\pi}\right)^2 = 450$ degrees.

compare: $E = \frac{450 - 314.56}{314.56} \times 100\%$

$= 43.06\%$

(b) $D_0 = \frac{4\pi}{\Omega_A} = \frac{4\pi}{0.09582} = 131.146$ (dimensionless).

$= 10\log_{10}(131.146)$ dB

$= 21.178$ dB

2.2 $\bar{E}(x,y,z,t) = Re\{\bar{E}(x,y,z)e^{j\omega t}\}$ (2-5)

$\bar{H}(x,y,z,t) = Re\{\bar{H}(x,y,z)e^{j\omega t}\}$ (2-6).

Rel $\bar{W} = \bar{E} \times \bar{H} = Re\{\bar{E} \cdot e^{j\omega t}\} \times Re\{\bar{H} e^{j\omega t}\}$

$= \frac{1}{2}\{\bar{E} \cdot e^{j\omega t} + \bar{E}^* \cdot e^{-j\omega t}\} \times \frac{1}{2}\{\bar{H} \cdot e^{j\omega t} + \bar{H}^* \cdot e^{-j\omega t}\}$.

$= \frac{1}{4}\{(\bar{E} \times \bar{H} e^{2j\omega t} + \bar{E}^* \times \bar{H}^* \cdot e^{2j\omega t}) +$

$(\bar{E} \times \bar{H}^* + \bar{E}^* \times \bar{H})\}$

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又因为 $\bar{E}\times\bar{H}e^{2j\omega t}+\bar{E}^*\times\bar{H}^*e^{2j\omega t}=2Re[\bar{E}\times\bar{H}e^{2j\omega t}]$

$\bar{E}\times\bar{H}^*+\bar{E}^*\times\bar{H}=2Re\{\bar{E}\times\bar{H}^*\}$

故 $\bar{W}=\frac{1}{2}Re[\bar{E}\times\bar{H}^*]+\frac{1}{2}Re[\bar{E}\times\bar{H}e^{2j\omega t}]$ 即(2-7)

得证.

2.3 (a) $\bar{W}_{rad}=\frac{1}{2}Re\{\bar{E}\times\bar{H}^*\}=\frac{E^2}{2\eta}\cdot\hat{a}_r=\frac{(5V/m)^2}{2\times120\pi}\cdot\hat{a}_r=0.03315\,\hat{a}_r\ W/m^2$

(b) $\bar{P}_{rad}=\oint_S W_{rad}\cdot ds$

$=\int_0^{2\pi}d\phi\int_0^{\pi}d\theta\sin\theta\, r^2\cdot W_{rad}$

$=2\pi\times(100m)^2\times\cos\theta\Big|_{\pi}^{0}\cdot0.03315\cdot\hat{a}_r$

$=4.16575\times10^3\ W.$

2.4 (a) $U(\theta)=\cos\theta$, $U(\theta)=\frac{1}{2}$时, $\theta'=\frac{\pi}{3}$, $U(\theta)=0$时, $\theta''=\frac{\pi}{2}$.

则 HPBW: $\theta_h=2\theta'=\frac{2\pi}{3}=120^\circ$

FNBW: $\theta_n=2\theta''=\pi=180^\circ$

(b) $U(\theta)=\cos^2\theta$; $U(\theta)=\frac{1}{2}$时, $\theta'=\frac{\pi}{4}$, $U(\theta)=0$时, $\theta''=\frac{\pi}{2}$

则 HPBW: $\theta_h=2\theta'=\frac{\pi}{2}=90^\circ$

FNBW: $\theta_n=2\theta''=\pi=180^\circ$

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(c). $U(\theta)=\cos(2\theta)$, $U(\theta)=\frac{1}{2}$时, $\theta'=\frac{\pi}{6}$, $U(\theta)=0$时, $\theta''=\frac{\pi}{4}$

则 HPBW: $\theta_h=2\theta'=\frac{\pi}{3}=60^\circ$

FNBW: $\theta_n=2\theta''=\frac{\pi}{2}=90^\circ$

(d). $U(\theta)=\cos^2(2\theta)$, $U(\theta)=\frac{1}{2}$时, $\theta'=\frac{\pi}{8}$, $U(\theta)=0$时, $\theta''=\frac{\pi}{4}$

则 HPBW: $\theta_h=2\theta'=\frac{\pi}{4}=45^\circ$

FNBW: $\theta_n=2\theta''=\frac{\pi}{2}=90^\circ$

(e) $U(\theta)=\cos(3\theta)$, $U(\theta)=\frac{1}{2}$时, $\theta'=\frac{\pi}{9}$, $U(\theta)=0$时, $\theta''=\frac{\pi}{6}$

则 HPBW: $\theta_h=2\theta'=\frac{2}{9}\pi=40^\circ$

FNBW: $\theta_n=2\theta''=\frac{\pi}{3}=60^\circ$

(f). $U(\theta)=\cos^2(3\theta)$, $U(\theta)=\frac{1}{2}$时, $\theta'=\frac{\pi}{12}$, $U(\theta)=0$时, $\theta''=\frac{\pi}{6}$

则 HPBW: $\theta_h=2\theta'=\frac{\pi}{6}=30^\circ$

FNBW: $\theta_n=2\theta''=\frac{\pi}{3}=60^\circ$

2.5 (a). $U(\theta)=\cos\theta\cos(2\theta)$. $\theta_1=0.481\pi$

$U(\theta)=\frac{1}{2}$时, $U(\theta)=0$时, $\theta_2=\frac{\pi}{4}$.

则 HPBW= $\theta_h=2\theta_1=55.584^\circ$

FNBW: $\theta_n=2\theta_2=\frac{\pi}{2}=90^\circ$

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(b). $U(\theta)=\cos^2\theta\cos^2(2\theta)$

$U(\theta)=\frac{1}{2}$时, $\theta_1=0.357\pi$. , $U(\theta)=0$时, $\theta_2=\frac{\pi}{4}$

则 HPBW: $\theta_h=2\theta_1=40.985^\circ$

FNBW: $\theta_n=2\theta_2=\frac{\pi}{2}=90^\circ$

(c) $U(\theta)=\cos\theta\cos(3\theta)$.

$U(\theta)=\frac{1}{2}$时. $\theta_1=0.337\pi$. , $U(\theta)=0$时. $\theta_2=\frac{\pi}{6}$

则 HPBW: $\theta_h=2\theta_1=38.668^\circ$

FNBW: $\theta_n=2\theta_2=\frac{\pi}{3}=60^\circ$

(d). $U(\theta)=\cos^2\theta\cos^2(3\theta)$

$U(\theta)=\frac{1}{2}$时. $\theta_1=0.251\pi$. , $U(\theta)=0$时, $\theta_2=\frac{\pi}{6}$

则 HPBW: $\theta_h=2\theta_1=28.745^\circ$

FNBW: $\theta_n=2\theta_2=\frac{\pi}{3}=60^\circ$

(e) $U(\theta)=\cos(2\theta)\cos(3\theta)$

$U(\theta)=\frac{1}{2}$时, $\theta_1=0.305\pi$; $U(\theta)=0$时. $\theta_2=\frac{\pi}{6}$

则 HPBW: $\theta_h=2\theta_1=34.942^\circ$

FNBW: $\theta_n=2\theta_2=\frac{\pi}{3}=60^\circ$

(f) $U(\theta)=\cos^2(2\theta)\cos^2(3\theta)$.

$U(\theta)=\frac{1}{2}$时. $\theta_1=0.223\pi$, $U(\theta)=0$时. $\theta_2=\frac{\pi}{6}$

则 HPBW: $\theta_h=2\theta_1=25.583^\circ$, FNBW: $\theta_n=2\theta_2=\frac{\pi}{3}=60^\circ$

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